

$$\mathbb{Z}_6$$



g

$$\mapsto \varphi_1(g)$$

$$\mapsto \varphi_2(g)$$

$$\mapsto \varphi_3(g)$$

g'

$$\mapsto \varphi'_1(g')$$



$$(g')^m = g$$

$$\varphi'_1((g')^m) = (\varphi'_1(g'))^m$$

$$\Rightarrow \varphi'_1(g) =$$

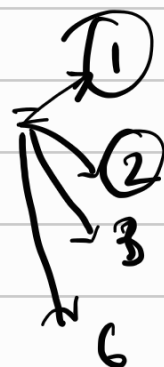
$$g^{-m} = g'$$



g_1, g_2 are generators $\mathbb{Z}_6$

then their images gives the same ^{set} homomorphism.

$$o(g) = 6$$



$$S_5$$

$$\mapsto \textcircled{1}$$

$$g \mapsto e \rightarrow \text{trivial } (1)$$

$$\mapsto \textcircled{2}$$

$$g_1 \mapsto \textcircled{2} \quad \left(\begin{smallmatrix} 5 \\ 2 \end{smallmatrix} \right) + \left(\begin{smallmatrix} 5 \\ 2 \end{smallmatrix} \right) \left(\begin{smallmatrix} 3 \\ 2 \end{smallmatrix} \right)$$

$$\mapsto \textcircled{3}$$

$$g_1 \mapsto \textcircled{3} \quad \left(\begin{smallmatrix} 5 \\ 3 \end{smallmatrix} \right)$$

$$\mapsto \textcircled{4}$$

$$g_1 \mapsto \textcircled{6} \quad \left(\begin{smallmatrix} 5 \\ 3 \end{smallmatrix} \right) \left(\begin{smallmatrix} 2 \\ 2 \end{smallmatrix} \right)$$

$$\{ \textcircled{1}, \textcircled{2}, 3, 4, 5 \}$$

$$\mapsto \textcircled{1} \textcircled{2}$$

$$\mapsto \textcircled{1} \textcircled{2} \textcircled{3} \textcircled{4}$$

